Assignment– Software Testing

Read the below requirement specification on Order Processing System for the organization called Material India Pvt. Ltd. (MIPL).

Answer the following using the standard template provided along with this assignment.

1. Go through the specification and

a)identify the Functional and Business flow of the App and state the flows, for eg **Login->Create order->save-> etc ,etc** .You are not expected to illustrate in a diagram. (B)Identify the Different Users of the app (20 marks)

2. Identify Functional and Quality attributes in the given app reqs. (5 Marks)

3. Design the scenarios in the Test Scenarios sheet. (20 marks)

4. Write test cases in the Test Cases sheet by providing Test Scenario reference. (20 marks)

5. Post defects in the Defects sheet by providing relevant details (20 marks)

6. Refer the last section on the change request, and identify the test scenarios and test cases impacted by the change request and write them down in the Impact Analysis sheet.(Change Request ID and Title, Description of the Change on modules;Test Scenarios,Testcases)

(15 marks)

Note: You need to only identify impacted test cases and write the extent of impact on the application.

General guidelines:

1. It is not expected of you to design every scenario and test cases in the whole application. Cover the breadth of the application while identifying scenarios, meaning to say all positive scenarios across the application, and then think of negative cases and exceptional cases for order processing work flow. That should suffice.

2. Focus should be on testing the workflow of the system and not the unit level testing.

3. Features marked as already tested and delivered must not be considered for either designing scenarios or writing test cases. Master setup functionalities are mentioned as already tested and delivered.

Project : Order Processing System Ver 1.0

1 Introduction

Material India Pvt. Ltd. (MIPL), is a mediator between IT companies and vendors of stationery items. IT companies place bulk stationery orders on a regular basis to MIPL and MIPL procures all the items in a particular order from multiple vendors. After processing the order, MIPL packages all the items in an order and delivers those stationeries to its customer who had placed the order. The current process is manual and has many issues. This has created the need for automation. Online payment by the customers and interaction with external vendors is not planned in either current or subsequent releases.

2 section A Details

The order processing system shall have

Master setup - Defining customers, Defining stationeries, and Defining users for employees of MIPL and customers of MIPL.

Transactions include placing an order by customer, order validation by order receiving team, order processing by order processing team, order delivery by order delivery team, order acknowledgement by customer

Reports - Generating reports by MIPL management team.

2.1 Master Setup

2.1.1 Defining Customers

The system shall make provision to define information about the customers in the system. The details about the customers shall include Customer abbreviation, customer name, customer address, phone number, email id, primary contact person, secondary contact person, credit limit. Edit and deactivate facility for the customer shall also be available. **The same has been tested and released to customer.**

2.1.2 Defining stationeries

The system shall make provision to define stationery items, which can be ordered by the customers. The details to be captured include stationery item abbreviation, stationery item name, and maximum no. of items that can be ordered at a time. Edit and deactivate facility for the customer shall also be available. **The same has been tested and released to customer.**

2.1.3 User creation

There shall be provision in the system to create users for employees of MIPL and also for its customers. The details to be captured for employees of the MIPL include User name, Password, Validity period and a flag whether to change the password at first login. In case of defining users

belonging to customers, the additional parameter customer name shall be selected as the customers are already defined. If the user is MIPL employee, then the department Order receiving / Order processing / Order delivery / Management must be selected. Login and password must be in between 8 – 12 characters inclusive. If flag to change password is selected, the system must force the user to change the password at first login. **The same has been tested and released to customer.**

2.2 Customer Ordering

In order to place the order online, the customer logs in with the user name and password. The details of the customer abbreviation, name, address, phone numbers shall be displayed on the screen. There shall be provision to change the above displayed details.

The details to be captured during placing the order by the customer include date of placing the order, stationery item, quantity of the stationery item, expected delivery date (by default, this date shall be 3 days from the current date), delivery address and phone number.

As the user enters the stationery item, the system shall show the price per unit quantity, and as the user enters the quantity of the stationery item, the system shall show the amount for the specified quantity.

It must be possible in the system to order multiple stationery items in a single order. When the order is placed, the status of the order shall be **"Submitted"**.

2.3 Order Validation

There shall be provision in the system to view the submitted orders and approve them as "Qualified" for order processing. There are a set of order receiving staff, whose primary responsibility is to view the received orders from its customers, and validate them. These order receivers log in into the system with the username and password provided to them.

The validation process includes checking the credit limit of the customer, who has placed the order, checking overdue amounts, availability of items in the market. If all the validations are successful, then these individuals mark the order as **"Qualified".**

Once the order is marked as qualified, the order is visible to the order-processing department.

2.4 Order Processing

The order processing team is responsible for collecting all the items in an order. A order processing team member logs into the system and views the orders in Qualified state.

There shall be provision in the system to mark individual item in an order as "**Received**". The status of the order shall be "**In Progress**" when at least one of the items in the order is received. The status "in progress" shall continue till all the items are received. Once all the items are received, then it must be possible to update the status to "**Completed**".

The order processing team puts delivery notes while passing the procured materials to the delivery departments. Hence, there shall be provision in the system to enter delivery notes. This shall be pure text content with notes up-to 10,000 characters. Once the delivery notes are provided, then the status of the order must be automatically marked as **"Ready for delivery".**

2.5 Order Delivery

There shall be facility in the system to view the orders, which are of the status “Ready for delivery”. The delivery team views the orders of this status along with the delivery notes, provided by the order processing team and make arrangements for actual delivery. Once all the items of a particular order are dispatched, the order delivery team must be able to change the

status of the order to **“Delivered”.** The delivery date shall be internally captured internally by the system.

3 Section B Requirements

3.1 Performance Requirements

PE-1: The system shall accommodate 80 users during the peak usage time, with estimated average session duration of 8 minutes.
PE-2: All Web pages generated by the system shall be fully downloadable in no more than 5 seconds over an 80KBps modem connection.
PE-3: Responses to queries shall take no longer than 5 seconds to load onto the screen after the user submits the query, while generating reports.
PE-4: The system shall display confirmation messages to users within 4 seconds after the user submits information to the system.

	Role	**Privileges**
3.2 Security Requirements #		
1	Administrator	To set up users, stationeries, and to define customers
2	Customer	To place order for stationeries through internet, Getting to know the status of the order they have placed, To know the payment details To provide comments/remarks about the delivery and quality of goods

3	Order Receiver	View all the orders placed by its customers Mark the order as “Qualified” in the system
4	Order Processor	View all the orders marked as “Qualified” Mark an individual item of the order as “Received” Mark the whole order as “Completed” Mark the whole order as “Ready for delivery”
5	Order Delivery	View all the orders with the status “Ready for delivery” Mark the order as “Delivered”
6	Manager	To view the status of all the orders in the system To generate all kinds of reports specified in the MIS reports section

4 Defect Scenarios

1. Users having logged in as Order receivers and Order processors and Order deliverers are also able to view the management reports in the system.

2. The system is not validating the order quantity value while the customer places the order against the Max. order Qty. set while defining stationeries in the master setup.

3. The order directly moves to Ready for delivery state, even though the delivery notes are not yet entered for the order, after procuring all items in the order.

4. While placing an order, the default customer address is displayed, but not allowed to edit and save.

5 Change Request

MIPL has experienced severe trouble sometime in delivering the items of an order within 3 days. That is why, in case of exceptional situations, MIPL has decided to introduce new statuses called “On hold - Qualified”, “On hold – In progress”, “On hold – Completed”, On hold – Ready for delivery”.

If the order is qualified and for some reason, it is not possible to procure the material immediately, the status will be moved “On hold – Qualified”.

If the order is under processing, some of the items are already received and there is a critical problem in procuring the rest of the items, the status shall be moved to “On hold – In progress”

If all the items in an order procured and the order is in completed state, there is some problem about how to package items, the status will be temporarily moved to “On hold – Completed”.

If all the items are procured and delivery notes are written, the order is in “ready for delivery” state, and there is an issue in delivering the items to the customer, then the status would be moved to “On hold – ready for delivery”.

These are additional variations from the normal flow of submitting, validating, processing and delivering an order. In each of the above cases, the order returns to the normal flow once the problem is resolved. For example, an order in the status “On hold – Completed”, returns to Ready for Delivery, once the packaging issue is resolved